

PAGO PAGO, American Samoa

WMO: 917650

Lat: 14.331S

Lon: 170.714W

Elev: 12

StdP: 14.69

Time Zone: -11.00 (W11)

Period: 94-19

WBAN: 61705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	73.7	74.9	65.6	94.5	79.2	67.6	101.5	79.4	27.1	79.4	25.5	79.9	4.9	300	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.7	90.8	81.1	89.8	80.8	89.1	80.7	83.4	87.6	82.7	87.0	82.2	86.6	11.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.4	168.9	86.1	81.6	164.4	85.6	80.9	160.6	85.2	47.4	87.8	46.7	87.3	46.0	86.8	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	22.5	20.8	DB	71.1	93.0	1.9	2.5	69.7	94.8	68.5	96.2	67.4	97.6	66.0	99.4	(4)
(5)			WB	66.1	84.6	2.0	1.4	64.7	85.6	63.5	86.4	62.4	87.2	61.0	88.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	83.1	83.3	83.3	83.0	82.2	81.6	80.9	80.9	81.7	82.1	82.7	83.0
	DBStd	2.05	1.91	2.03	1.80	1.81	1.87	1.74	1.90	1.81	1.84	1.90	1.91	1.84
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11792	1027	932	1033	991	997	948	958	957	952	994	980	1023
	CDD65	6317	562	512	568	541	532	498	493	492	502	529	530	558
	CDH74	71316	6698	6129	6822	6431	5951	5378	5009	4953	5361	5832	6162	6590
CDH80	23191	2528	2376	2654	2416	1884	1427	1107	1099	1435	1740	2129	2396	
Wind	WSAvg	10.9	8.9	9.0	8.0	8.7	10.5	13.2	13.4	13.2	13.7	11.9	10.2	9.6
Precipitation	PrecAvg	117.7	13.3	12.4	11.4	11.5	10.3	6.5	6.0	6.6	7.3	10.6	10.9	13.4
	PrecMax	153.1	28.0	32.7	25.6	28.3	29.1	12.6	14.4	18.5	25.3	29.1	25.7	27.0
	PrecMin	59.8	4.6	2.0	4.0	0.7	1.6	2.6	0.7	0.4	1.0	2.1	3.2	2.2
	PrecStd	20.4	6.0	6.1	5.4	6.6	6.3	2.9	3.4	4.5	5.4	5.7	5.0	6.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.4	92.0	91.6	91.0	89.3	87.9	87.6	87.7	88.9	89.8	91.1	91.1
		MCWB	80.9	81.8	81.3	81.2	80.2	79.8	79.2	79.3	79.7	80.7	81.1	81.1
	2%	DB	89.8	90.0	89.9	89.4	88.0	86.8	86.3	86.4	87.5	88.1	89.2	89.5
		MCWB	80.7	81.0	81.1	81.0	79.9	79.4	78.7	78.7	79.5	80.4	80.8	80.5
	5%	DB	88.9	89.1	89.1	88.4	87.1	86.0	85.5	85.6	86.4	87.0	88.0	88.3
		MCWB	80.6	80.7	80.9	80.6	79.6	79.1	78.4	78.4	79.1	79.9	80.6	80.4
	10%	DB	87.8	88.0	88.0	87.5	86.1	85.1	84.5	84.5	85.5	86.0	87.1	87.5
		MCWB	80.4	80.4	80.6	80.3	79.4	78.7	78.0	78.0	78.7	79.3	80.2	80.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.6	83.9	84.3	83.7	82.7	81.9	81.9	82.2	82.5	83.3	83.5	83.7
		MCDB	88.1	88.6	88.1	87.5	86.6	85.3	84.6	85.0	86.4	87.0	87.6	87.4
	2%	WB	82.6	82.8	82.7	82.7	81.7	80.8	80.5	80.5	81.2	81.8	82.3	82.5
		MCDB	87.3	87.5	87.5	86.8	85.7	84.7	83.9	83.6	85.3	85.7	86.8	86.7
	5%	WB	81.8	82.0	81.9	81.9	80.9	80.1	79.7	79.7	80.2	80.9	81.5	81.7
		MCDB	86.7	86.8	87.0	86.3	85.0	84.1	83.4	83.2	84.4	85.1	86.2	86.2
	10%	WB	81.0	81.2	81.2	81.2	80.3	79.6	79.0	79.0	79.6	80.1	80.8	80.9
		MCDB	86.2	86.3	86.5	85.9	84.4	83.7	83.0	82.9	83.7	84.4	85.6	85.7
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.3	9.7	9.1	7.9	6.9	6.9	7.2	7.7	8.0	8.4	8.8
		MCDBR	10.4	10.5	10.5	9.9	9.0	7.9	8.1	8.4	8.7	9.0	9.5	9.9
		MCWBR	5.0	4.7	5.0	4.9	4.7	4.3	4.5	4.7	4.7	4.8	4.8	4.9
	5% WB	MCDBR	9.1	9.1	9.6	9.0	8.4	7.5	7.8	7.9	8.3	8.7	9.0	9.2
		MCWBR	5.1	5.0	5.1	5.3	5.0	4.5	4.9	4.9	5.0	5.2	5.2	5.3
Clear-Sky Solar Irradiance	taub	0.387	0.378	0.374	0.382	0.389	0.392	0.392	0.397	0.399	0.386	0.383	0.385	
	taud	2.633	2.670	2.686	2.656	2.618	2.590	2.584	2.562	2.562	2.609	2.637	2.635	
	Ebn at Noon	303	304	298	286	273	267	270	277	287	298	303	304	
	Edh at Noon	32	31	30	29	28	28	29	31	33	32	32	32	
All-Sky Solar Radiation	RadAvg	1874	1864	1857	1639	1443	1380	1428	1654	1855	1935	1905	1886	
	RadStd	188	164	138	137	102	68	56	77	106	103	92	103	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page